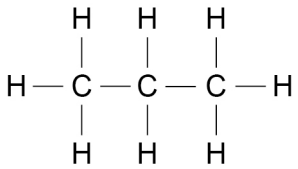
Do not write outside the box

0 5 This question is about hydrocarbons.

Figure 9 shows a hydrocarbon.

Figure 9



0 5 . 1 Complete the formula for the hydrocarbon shown in **Figure 9**.

[1 mark]

C_____H_____

0 5 . 2 What is the name of the hydrocarbon in **Figure 9**?

[1 mark]

_____

0 5 . 3 Which homologous series does the hydrocarbon in **Figure 9** belong to?

[1 mark]

_____

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0 5 . 4 30 g of another hydrocarbon contains 24 g of carbon.

Which calculation gives the percentage of carbon in the hydrocarbon?

[1 mark]

Tick (✓) **one** box.

$\frac{24 \times 30}{100}$ ☐

$\frac{100 \times 30}{24}$ ☐

$\frac{24 \times 100}{30}$ ☐

$\frac{24}{30 \times 100}$ ☐

0 5 . 5 **Table 3** shows boiling points of some hydrocarbons.

Table 3

Formula of hydrocarbon	Boiling point in °C
C_2H_6	–89
C_4H_{10}	0
C_6H_{14}	69
C_8H_{18}	125
$C_{10}H_{22}$	174

Describe how the boiling points change as the number of carbon atoms in the hydrocarbon increases.

[1 mark]

_____

_____

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Hydrocarbons can be cracked.

0 5 . 6 Give **one** condition used to crack hydrocarbons.

[1 mark]

_____

_____

0 5 . 7 Balance the equation for the cracking of C_6H_{14}

[1 mark]

$$C_6H_{14} \rightarrow C_2H_6 + _____\ C_2H_4$$

0 5 . 8 Give **one** reason why hydrocarbons are cracked.

[1 mark]

_____

_____

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0 5 . 9 Window frames can be manufactured from wood or plastic.

Table 4 shows the results of a life cycle assessment (LCA) for making one wooden and one plastic window frame.

Both window frames are the same size.

Table 4

Table 4 not reproduced here due to third-party copyright restrictions

Give **three** advantages of using wood instead of plastic in the manufacture of window frames.

[3 marks]

Advantage of wood 1 _____

_____

_____

Advantage of wood 2 _____

_____

_____

Advantage of wood 3 _____

_____

_____

11

Turn over for the next question